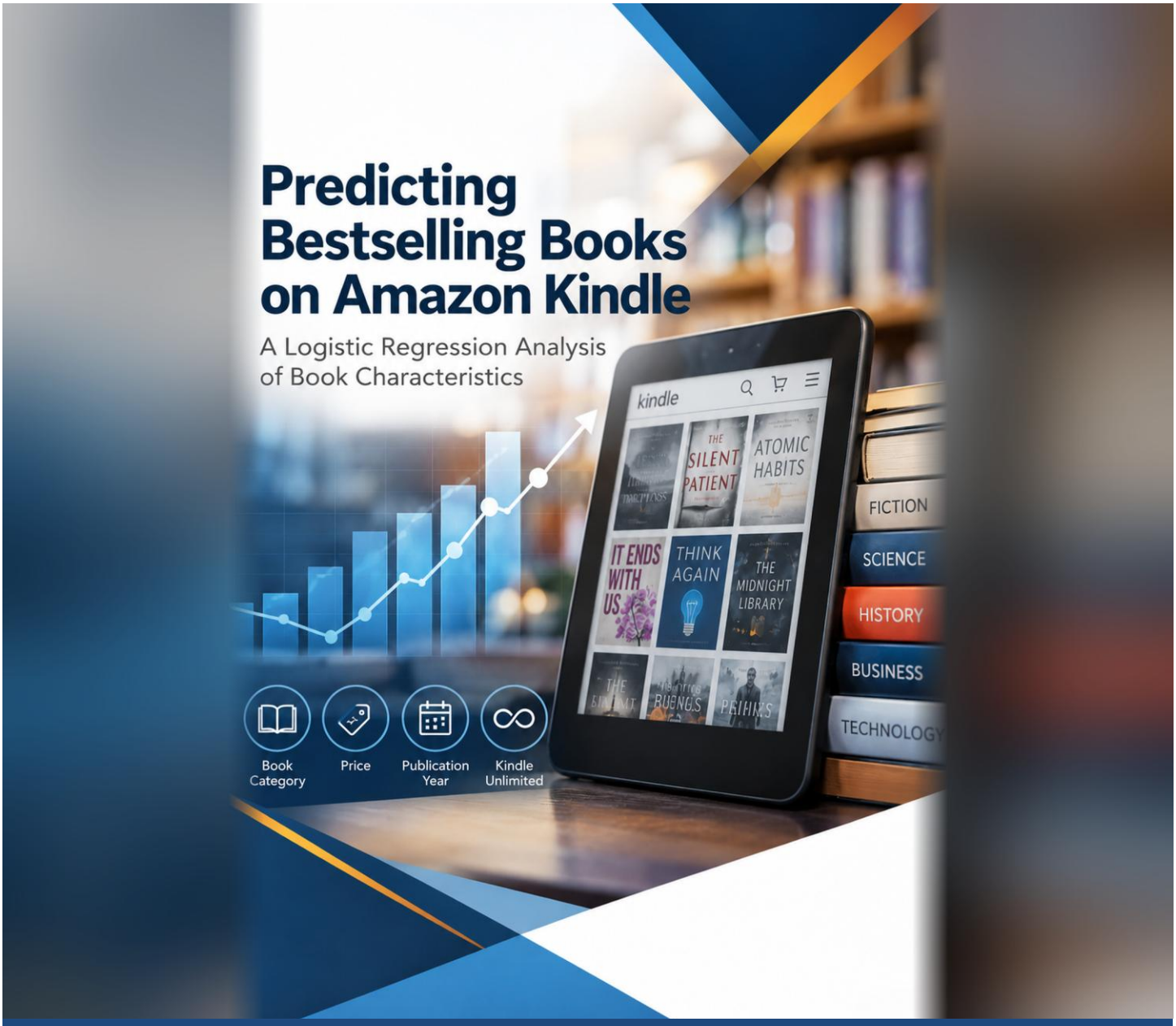


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P.O.E Part 2

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Introduction and research question

The publisher wants to identify books that could become bestsellers. I will investigate this using Logistic Regression, with a public dataset of Amazon Kindle listings. Since the available target describes bestseller status at the time of collection, the model will classify observed designations rather than verify future commercial success.

Dataset selection and suitability

I selected the Amazon Kindle Books Dataset 2023 because it provides book characteristics alongside an existing binary bestseller label (asaniczka, 2023). Spark confirmed 133,102 listings, 16 fields and correctly parsed numerical, Boolean and date values after I adjusted the CSV quote-handling option. The dataset comfortably exceeds the 10,000-record requirement.

Of these books, 2,236 (1.68%) carry the bestseller label and 130,866 (98.32%) do not. This imbalance is a major challenge: predicting every book as a non-bestseller would already achieve 98.32% accuracy while finding none of the books the publisher is interested in. Saito and Rehmsmeier (2015) explain why precision-recall analysis is especially helpful when positive cases are rare. I will therefore assess more than accuracy when evaluating the model.

The data-quality checks found missing publication dates for 49,016 books (36.83%), seller details for 9,233 (6.94%) and author names for 425 (0.32%). All ASINs are unique, and no rows are exact duplicates, although different editions of a book may still appear under different identifiers. I also found 64,670 zero review counts (48.59%), 4,066 zero prices (3.05%), 3,182 zero-star ratings (2.39%) and one publication date after 2023. No prices or review counts were negative, and no ratings fell outside 0–5. The zeros may represent unavailable information rather than genuine zero values, so I will not treat them as errors without further evidence.

The dataset is suitable for investigating binary Logistic Regression because it has a defined target and book-level predictors such as category, price and Kindle Unlimited availability. However, suitability depends on when the publisher would know each predictor. Reviews, star ratings and editorial or Goodreads endorsements may reflect later popularity, making them unsuitable for an honest pre-publication prediction. As Kaufman et al. (2012) explain, using information unavailable at prediction time introduces data leakage. I will examine this risk when selecting features.

Overall, the dataset is large enough and sufficiently structured for the assignment, but it is not evidence of future sales outcomes. The analysis will be framed as a retrospective classification of an Amazon bestseller designation, with that distinction kept clear when interpreting its usefulness to the publisher.

Analysis plan

The aim of this analysis is to develop and evaluate a Logistic Regression model that can help the publisher identify books associated with bestseller status. My initial dataset evaluation identified two important challenges: the relatively small number of bestsellers and the possibility that some variables reflect a book's popularity after publication. These findings have guided my approach to exploring the data, selecting features, training the model and evaluating its performance.

Exploratory Data Analysis

I will use Apache Spark to examine how bestseller status varies across the available book characteristics, particularly categories, prices and publication dates. I will also investigate missing information and unusual values that could affect the reliability of the analysis. Rather than producing graphs for every variable, I will focus on patterns that can inform feature selection and model training.

Feature Selection

I will select features based on their relevance to the publisher's question, their data quality and whether the information could reasonably be available when assessing a book. According to Kaufman, et al. (2012), data leakage can occur when a model learns from information that would not legitimately be available during prediction. For this reason, customer reviews, ratings and editorial endorsements will not be included in the primary model. The final selection of features will follow the EDA findings.

Model Training

The Logistic Regression algorithm will be implemented from scratch using Python and NumPy. Hastie, Tibshirani and Friedman (2009) explain the statistical principles underlying Logistic Regression and regularization, which provide a foundation for the implementation. I will use binary cross-entropy to measure prediction error and gradient descent to update the model's coefficients.

A baseline model will be trained first, followed by experiments with L2 regularization and class weighting. These experiments will help me assess whether the model can generalize effectively while identifying more of the relatively rare bestseller observations.

Model Evaluation

I plan to divide the dataset into training, validation and test sets, using approximately 70%, 15% and 15% of the observations respectively. Where possible, related editions will be kept within the same partition to reduce the risk of information leakage. All preprocessing parameters will be learned from the training data.

Given that only 1.68% of the observations are bestsellers, accuracy alone would provide a misleading assessment. He and Garcia (2009) discuss the difficulties associated with imbalanced learning, while Saito and Rehmsmeier (2015) demonstrate why precision-recall analysis is especially informative when positive cases are rare. I will therefore use precision, recall, F1-score and average precision as important evaluation measures, supported by ROC-AUC and a confusion matrix.

The validation set will be used to compare training approaches and select an appropriate classification threshold. The test set will be reserved for the final evaluation.

Reporting

The final report will explain the findings from the EDA, the data preparation decisions, how the Logistic Regression algorithm was implemented and the results of the model evaluation. I will compare the initial and improved models and discuss what their performance means for the publisher.

I will also acknowledge that the dataset records Amazon bestseller status rather than future sales performance. Any recommendations will therefore be limited to what the results can reasonably support.

Exploratory data analysis

I used Apache Spark to explore which book characteristics might help the publisher distinguish bestsellers from other titles. Rather than examining every available variable, I focused on features that could inform the Logistic Regression model and investigated whether their relationships with bestseller status were reliable enough to consider.

Preparing the Data for Analysis

Before examining relationships with bestseller status, I identified 1,019 groups containing repeated title-author combinations, covering 2,063 listings. Although these were not confirmed duplicates, they could represent different editions of the same book.

I therefore divided the dataset into training, validation and test sets while keeping potentially related listings together. The training set contained 93,390 books, including 1,571 bestsellers (1.68%). No title-author group appeared in more than one partition.

This allowed me to conduct the remaining EDA using training data only, keeping the validation and test sets separate for model evaluation.

Book Categories and Bestseller Status

Yucesoy et al. (2018) found that bestseller representation differed across book genres in their study of US hardcover books. This provided a useful reason to investigate whether category was also associated with bestseller status in the Kindle dataset.

The training data contained 31 categories. Among the largest categories, Science & Math had a bestseller rate of 3.05%, followed by Teen & Young Adult (2.73%) and Children's eBooks (2.63%). These were higher than the overall training rate of 1.68%.

The differences suggest that category may contain useful information for classification. However, the findings describe associations within this dataset and do not establish that publishing in a particular category increases a book's chances of success.

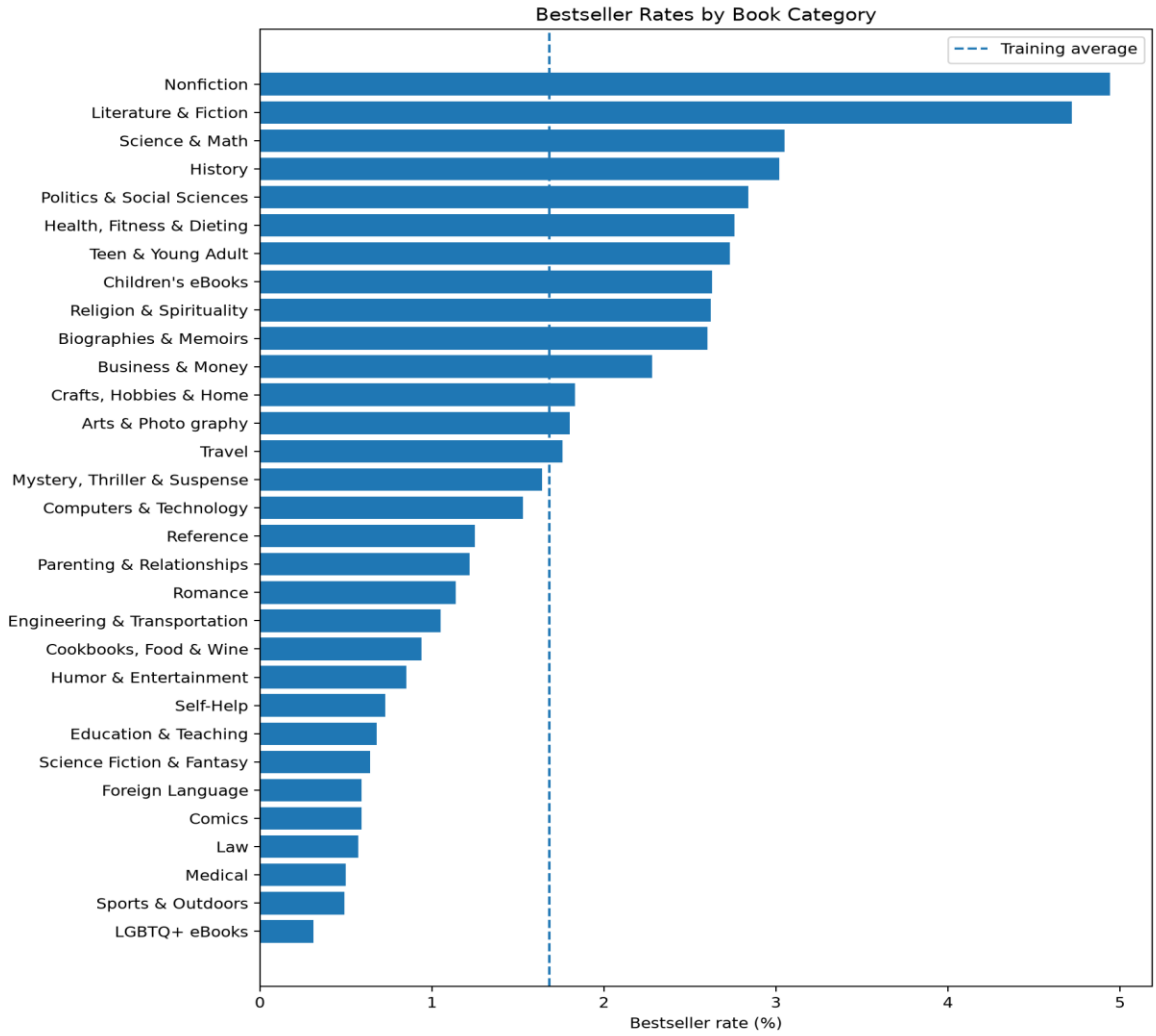


Figure 1 - Bestseller rates across Amazon Kindle book categories in the training dataset.

Book Prices and Bestseller Status

The price analysis showed that bestsellers generally had lower recorded prices. Their median price was 5.99, compared with 9.99 for non-bestsellers. A similar difference appeared at the 90th percentile, suggesting that the pattern was not simply caused by a few unusually expensive books.

Schmidt-Stölting, Blömeke and Clement (2011) demonstrate that the factors associated with commercial book success can vary across publishing contexts and formats. This is important when interpreting our results because the dataset records marketplace prices rather than verified launch prices.

I therefore identified price as a potentially useful predictor, while recognizing that the observed relationship may also reflect promotions, categories or changes after publication.

Publication Dates

Among books with recorded publication dates, bestsellers had a median publication year of 2021, compared with 2019 for non-bestsellers.

However, publication dates were missing for 39.53% of bestsellers and 36.88% of non-bestsellers. These substantial gaps limit how confidently the observed pattern can be applied to the full dataset.

Publication year remains a candidate feature, but the missing information will require careful treatment before training.

Kindle Unlimited Availability

Kindle Unlimited showed a particularly noticeable relationship with bestseller status.

Books available through the Programme had a bestseller rate of 4.10%, compared with 0.79% for other books. Although Kindle Unlimited books represented approximately 27% of the training data, they accounted for 65.8% of its bestsellers.

According to Amazon Kindle Direct Publishing (n.d.), Kindle Unlimited participation is linked to KDP Select enrolment, which can occur before or after publication. Consequently, the relationship observed in this dataset does not establish that Kindle Unlimited participation causes commercial success.

Nevertheless, the strength of the association makes it worth investigating during modelling, provided its timing limitations are acknowledged.



Figure 2 - Bestseller rates by Kindle Unlimited availability compared with the overall training average.

Implications for Model Development

The EDA identified category, price, publication year and Kindle Unlimited availability as candidate predictors. These findings provide a basis for feature selection, but their actual contribution to prediction must still be evaluated.

Kuhn and Johnson (2019) explain that feature engineering and selection should be guided by the modelling problem and assessed in relation to predictive performance. Following this approach, I will distinguish between information that could be available to a publisher before release and information that may only be available later.

Customer reviews, ratings and editorial endorsements will therefore be excluded from the primary model because they may reflect a book's existing popularity.

The analysis also confirmed the importance of accounting for class imbalance. With only 1.68% of training records labelled as bestsellers, the model will need to be evaluated using measures that reveal how effectively it identifies the minority class.

Overall, the EDA has provided a clear direction for feature preparation and Logistic Regression training, while highlighting limitations that must be considered when interpreting the final results.

Feature Selection and Preparation

The EDA helped me identify four characteristics worth investigating: book category, price, publication year and Kindle Unlimited availability. I selected these because they showed differences between bestsellers and non-bestsellers, while remaining relatively straightforward to interpret.

Kuhn and Johnson (2019) explain that feature selection should reflect the purpose of a predictive model rather than simply include every available variable. Following this approach, I excluded identifiers, URLs and high-cardinality text fields. I also excluded reviews, ratings and editorial endorsements because they may reflect popularity after publication. As Kaufman, et al. (2012) explain, information unavailable at the intended prediction point can produce misleading model performance.

Handling Missing Values

Of the selected features, only publication year contained missing values, affecting 34,486 training records (36.93%). Removing these observations would have discarded a substantial portion of the data.

I therefore used median imputation to replace missing publication years and created a separate indicator to distinguish records with unavailable dates. This preserved the observations without treating the estimated years as genuine publication dates.

Preparing Numerical and Categorical Features

I applied one-hot encoding to the 31 book categories because they represent distinct groups rather than numerical rankings.

For price, I used a logarithmic transformation to reduce the influence of unusually high values. Price and publication year were then standardized to place them on comparable scales, supporting more stable optimization during gradient descent.

Kindle Unlimited availability and the missing-publication-year indicator were retained as binary features.

To prevent preprocessing leakage, I fitted the transformations using the training data only and applied them unchanged to the validation and test sets.

Final Modelling Data

The resulting datasets contained 34 numerical features each. The training set contained 93,390 observations, the validation set 20,028 and the test set 19,684. All features were successfully converted into numerical values, with no missing or invalid values remaining.

These preparations provided a consistent foundation for implementing Logistic Regression from scratch.

Although recorded price and Kindle Unlimited availability may not represent their values before publication, they are retained for the initial classification of bestseller status at collection time. This limitation will be considered when interpreting the final model's usefulness to the publisher.

Logistic Regression implementation

To address the publisher's classification problem, I implemented Logistic Regression from scratch using Python and NumPy. Rather than relying on an existing classifier, I developed the calculations needed to estimate bestseller probabilities, measure prediction errors and update the model's coefficients.

Model Design

Logistic Regression uses a weighted combination of input features to estimate the probability of a binary outcome. Bishop (2006) explains that the sigmoid function converts this weighted score into a value between zero and one.

In my implementation, the predicted probability represents the likelihood that a book belongs to the bestseller class, based on its selected characteristics.

I used binary cross-entropy to measure prediction error and calculated the gradients needed to update the coefficients. These operations were implemented using NumPy's vectorized calculations to avoid processing observations individually.

I also used a numerically stable loss calculation to reduce the risk of errors when predicted probabilities approach zero or one.

Training with Gradient Descent

I implemented full-batch gradient descent to minimize the loss function. At each iteration, the algorithm calculates the prediction error, determines the gradients and adjusts the coefficients according to the learning rate.

Hastie, Tibshirani and Friedman (2009) provide the statistical foundation for Logistic Regression and regularized model fitting.

For the initial model, I used a learning rate of 0.2 and a maximum of 1,000 iterations. I also included optional L2 regularization and class weighting in the implementation, allowing these methods to be investigated later without rewriting the algorithm.

The initial model was trained without either adjustment to establish a reference for evaluating improvements.

Verifying the Implementation

Before training, I tested the analytical gradients against numerical approximations using SciPy's gradient-checking function.

The test returned an error of 3.70×10^{-8}, indicating very close agreement between the two calculations. I conducted this check with both regularization and unequal class weights enabled to verify the extended loss function.

The initial training reduced binary cross-entropy from 0.085398 to 0.079021, with the loss decreasing consistently throughout all 1,000 iterations.

The final gradient norm was 0.003297, while the loss improved by a further 0.000228 during the final 100 iterations. Although these findings confirm stable learning, they do not establish complete convergence because training reached the maximum iteration limit.

I therefore retained this implementation as the initial model. The next stage will evaluate its ability to identify bestsellers using data that was not used for training and investigate whether further adjustments improve its performance.

Model evaluation and improvements

I evaluated the Logistic Regression model to determine whether it could identify bestsellers in data that had not been used for training. Since only 1.68% of the original dataset consisted of bestsellers, I placed particular emphasis on precision, recall, F1-score and average precision rather than accuracy alone.

Saito and Rehmsmeier (2015) explain that precision-recall analysis is particularly informative for imbalanced classification problems. This was especially relevant to my analysis because a model could achieve high accuracy simply by predicting that every book was a non-bestseller.

Initial Model Performance

The initial model achieved 98.37% accuracy on the validation set but failed to identify any of the 327 actual bestsellers at the default threshold of 0.5.

However, its average precision of 0.0469 and ROC-AUC of 0.7370 suggested that it had learned some ability to distinguish between the classes.

I therefore investigated whether changing the classification threshold could improve its usefulness.

Model Improvement

Lowering the threshold to 0.0505 allowed the model to identify 35 bestsellers, improving recall to 10.70% and F1-score to 0.1090.

I then tested class weighting to give greater importance to the relatively rare bestseller observations. He and Garcia (2009) discuss cost-sensitive learning as one approach to the challenges of imbalanced classification.

Class weighting improved average precision from 0.0469 to 0.0689 and increased the best validation F1-score to 0.1409. The weighted model correctly identified 47 bestsellers, although it also incorrectly flagged 293 non-bestsellers.

I subsequently tested L2 regularization and additional training iterations. Stronger regularization reduced coefficient size but did not improve average precision. Training for 2,000 iterations produced only small changes in performance.

Based on the validation results, I selected the class-weighted model trained for 1,000 iterations without L2 regularization. Its classification threshold of 0.8632 was selected using the validation set.

Independent Test Results

I evaluated the selected model on the previously untouched test set, containing 19,684 books and 338 bestsellers.

The model achieved 97.01% accuracy, 0.0665 average precision and 0.8039 ROC-AUC. Its bestseller precision was 12.08%, recall was 11.83%, and F1-score was approximately 0.1196.

The confusion matrix showed that the model correctly identified 40 bestsellers while missing 298. It also incorrectly flagged 291 non-bestsellers.

Although the model's accuracy was lower than the 98.28% majority-class baseline, it successfully identified some bestsellers that the baseline would have missed. Its average precision also remained reasonably close to the validation result, suggesting that its ranking performance was relatively consistent.

Nevertheless, the model still missed most bestsellers, and only approximately 12% of the books it flagged were actually labelled as bestsellers. Figure 3 provides a clearer view of the model's ranking performance and the classification errors observed on the independent test set.

Table 1 - Final Logistic Regression Performance

Evaluation measure	Result
Books evaluated	19,684
Actual bestsellers	338
Books flagged for review	331
Correctly identified bestsellers	40
Incorrectly flagged books	291
Bestseller precision	12.08%
Bestseller recall	11.83%
Average precision	0.0665
ROC-AUC	0.8039

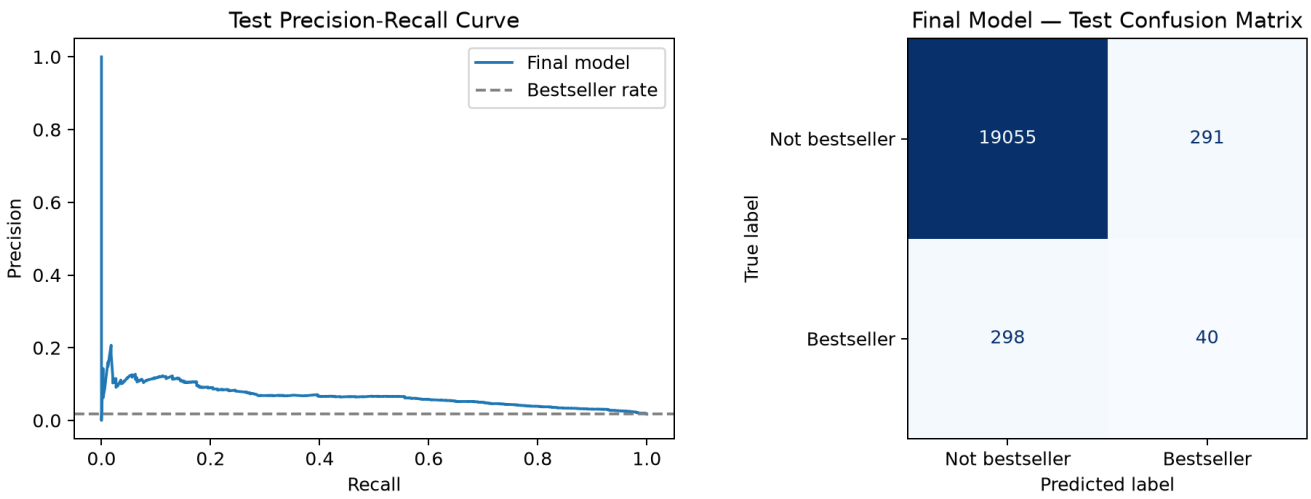


Figure 3 - Final Logistic Regression performance on the independent test set, showing the precision-recall curve and confusion matrix.

Conclusions and limitations

The aim of this investigation was to determine whether Logistic Regression could help a publisher identify potential bestsellers. I implemented the algorithm from scratch using Python and NumPy, explored the available book characteristics and tested several ways of improving its performance. The analysis revealed meaningful associations between bestseller status and features such as category, price, publication year and Kindle Unlimited availability. Class weighting produced the clearest improvement, increasing validation average precision from 0.0469 to 0.0689.

On the independent test set, the final model achieved an average precision of 0.0665 and ROC-AUC of 0.8039. However, it correctly identified only 40 of the 338 bestsellers, while missing 298 and incorrectly flagging another 291 books. Although the model demonstrated some ability to distinguish bestsellers from other titles, its precision and recall remain too limited for dependable publishing decisions. At best, it could help the publisher narrow down books for further investigation rather than determine which titles are likely to succeed.

The dataset also presents important limitations. Bestseller status reflects Amazon's designation at collection time rather than verified future sales. Some predictors, particularly price and Kindle Unlimited availability, may have changed after publication. Missing publication dates and the considerable imbalance between the two classes further limited the analysis. Consequently, the results demonstrate associations within the available data rather than the ability to forecast future commercial success.

A more reliable approach would require historical sales records, book characteristics captured before publication and evaluation across different release periods. This would provide a stronger basis for identifying promising titles before they become successful.

Overall, the project demonstrates a working and independently evaluated Logistic Regression implementation, while also showing why a technically successful model is not necessarily ready for practical use. The findings provide a useful starting point, but better data and further validation would be necessary before the publisher could rely on its predictions.

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